

# Vol 2 Chapter 12 — The Experimental Program: Substrate Engineering Falsifiers

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## Chapter 12 — The Experimental Program: Substrate Engineering Falsifiers

*Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine\_DIV5\_QM\_GR\_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.*

A framework is only as strong as its falsifiability. BST's experimental program — the SP-29 Casimir Mechanism Investigation and the SP-30 Substrate Engineering Program — collectively target the substrate's positive predictions for laboratory verification across a suite of experiments at 80K to 500K scales.

This chapter sets out the program's experimental targets, the substrate-mechanism predictions each test, and the per-experiment cost and timeline estimates. The program is what makes BST a falsifiable scientific framework rather than a theoretical exercise.

### 12.1 The substrate-engineering framework

The substrate is not directly probable at the Koons-tick scale ( $10^{-120}$  seconds) with current or foreseeable instruments. But the substrate's commitment-cycle output produces predictions at *observable* scales, and those predictions admit experimental tests.

The substrate-engineering framework — Casey's framing from May 2026 — is the methodology of designing experiments that probe substrate-derived predictions at scales where current technology can measure them. The SP-29 and SP-30 programs collectively target a suite of such experiments.

## 12.2 SP-29 Casimir Mechanism Investigation

The substrate's vacuum produces both the laboratory Casimir effect and the cosmological constant (Volume 4 Chapter 4  $\Lambda$ -Casimir unification, Lyra T2418). SP-29 targets precision Casimir measurements that test substrate-specific corrections:

- H4 Cs-137 measurement: precision Casimir-force measurement using Cs-137 sources, testing the substrate's predicted Casimir corrections at the 0.1% level
- Cost: 60-90K
- Timeline: 6-12 months from procurement to first data
- Falsifier sharpness: HIGH (precision regime accessible)

## 12.3 SP-30 Substrate Engineering Program

The SP-30 program covers five experimental designs targeting different substrate predictions:

- SP-30-1 Precision Bell experiment (300-500K): tests the substrate sub-Tsirelson signature  $|S|^2 \leq 126/16$  with the gap  $1/8 = 1/2^{N_c}$  as the substrate-specific prediction. Outreach targets: Vienna, Caltech, Munich, Hanson. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015):  $S = 2.82759 \pm 0.00051$ ,  $41.9\sigma$  above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]
- SP-30-2 Eigentone signal (200K): tests the substrate's predicted natural-frequency signatures via NIST/PTB/JILA/MPI atomic-clock infrastructure.
- SP-30-3 Commitment manipulation (80-150K): tests the substrate's predicted quantum-eraser revival amplitude correction at  $1/N_{\max} \approx 0.73\%$  via precision delayed-choice quantum eraser. Filed paper-grade May 23, 2026.
- SP-30-4 Time-granularity measurement (200-400K): tests the substrate clock cycle  $N_c \cdot t_{\text{Planck}}$  via next-generation optical lattice clocks at  $10^{-19}$  precision.
- SP-30-5 Substrate parallelism architecture: theoretical-first development; experimental design follows. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015):  $S = 2.82759 \pm 0.00051$ ,  $41.9\sigma$  above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

Per-experiment timelines are 6-12 months from procurement; total program cost  $\sim 640$ -900K.

## 12.4 The Five-Absence falsifiers

In parallel with the substrate-engineering positive-prediction tests, the Five-Absence framework (Chapter 11) provides experimental falsifiers via:

- Proton-decay searches (Super-Kamiokande, Hyper-Kamiokande)
- Magnetic-monopole searches (MoEDAL, IceCube)
- Sterile-neutrino searches (LSND/MiniBooNE/IceCube)
- Supersymmetric-particle searches (LHC, future colliders)
- Grand-unified-theory signatures (precision SM tests, proton decay)

All five experimental programs are currently running. BST predicts they will continue to find nothing positive; current results are consistent with this prediction.

## 12.5 The CPT falsifier

CPT violation at any precision would refute the substrate's  $SO_0(5, 2)$  structure (Volume 0 Chapter 8, structural CPT). Current experimental precision:  $\sim 10^{-18}$  on proton/antiproton mass differences. BST predicts no CPT violation at any precision; the test will tighten as experimental precision improves.

## 12.6 What comes next

This is the final chapter of Volume 2. After Volume 2, the curriculum continues with:

- Volume 3 — Nuclear and Atomic Physics
- Volume 4 — General Relativity and Cosmology (Newton's G and the cosmological constant already rewritten as recruiter chapters)
- Volume 5 — Quantum Mechanics
- Volume 6 — Statistical Mechanics and Thermodynamics
- Volume 7 — Electromagnetism
- Volume 8 — Classical Mechanics
- Volume 9 — Condensed Matter Physics
- Volume 10 — Mathematical Methods
- Volume 11 — Generative Geometry and Topology
- Volume 12 — Chemistry
- Volume 13 — Biology
- Volume 14 — Information Theory
- Volume 15 — Methodology

The substrate framework continues to derive observables across every physics domain. Each volume applies the same substrate-mechanism pattern to the relevant physical regime.

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Where to look this up: SP-29: `notes/SP29-1_H4-Cs137_paper_grade_proposal.md`.  
SP-30 series: per-sub-item paper-grade proposals (SP-30-1 through SP-30-5 filed).

Five-Absence Set: notes/Five\_Absence\_Predictions\_Principle.md. For experimental status of the falsifiers: Particle Data Group annual reviews; relevant experiment collaboration publications. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015):  $S = 2.82759 \pm 0.00051$ ,  $41.9\sigma$  above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]